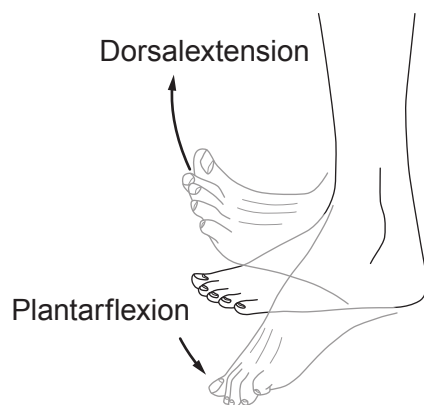


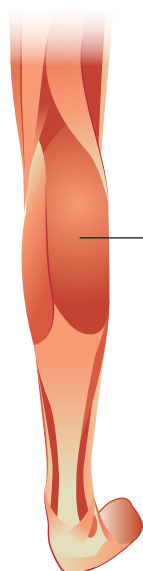
3. Healthy muscles and joints are essential for pain-free movement. These are very important for successful athletes.

- (a) The ankle is a type of hinge joint that is used when we walk and run. Its action is shown in the diagram below.



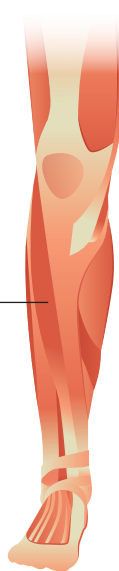
This movement is controlled by an antagonistic pair of muscles called the calf and tibialis anterior. The calf is at the back of the leg and the tibialis anterior is at the front of the leg.

**Back of leg**



Calf  
muscle

**Front of leg**



Tibialis  
anterior  
muscle

Explain how this pair of muscles produce dorsalextension and plantarflexion.

[3]

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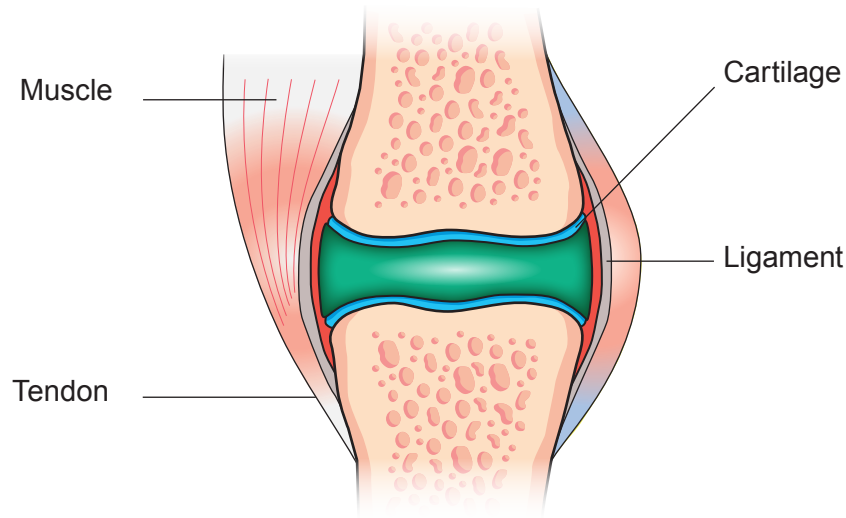
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- (b) The knee is a synovial hinge joint.



Explain how damage to cartilage and ligaments may affect an athlete.

[3]

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- (c) Two athletes, Dafydd and Tom, compete in a 400 m training race and their finish times are identical. Their times to travel certain distances within the race are measured and are shown in the table below.

| Distance (m)      | 0 | 50   | 100   | 150   | 200   | 300   | 400   |
|-------------------|---|------|-------|-------|-------|-------|-------|
| Tom's time (s)    | 0 | 7.65 | 13.65 | 20.48 | 27.30 | 41.93 | 56.55 |
| Dafydd's time (s) | 0 | 6.10 | 11.40 | 18.40 | 25.40 | 40.95 | 56.55 |

- (i) Dafydd thought his fastest speed was for the last 100 m of the race. Explain whether you agree.

[3]

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- (ii) Since they finished the race in the same time, Tom thought their maximum speed must have been the same. Plot graphs of distance against time for Tom and Dafydd and explain whether you agree with Tom. [5]



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